

CHE 311H - Fall 2023
Homework #4

Instructions: Show all work and box in your final answer. Write all group member's names on the first page and the group number in top corner of all subsequent pages. Remember, illegible solutions will receive no credit. Additionally, regrade requests must be submitted within 2 weeks of the due date.

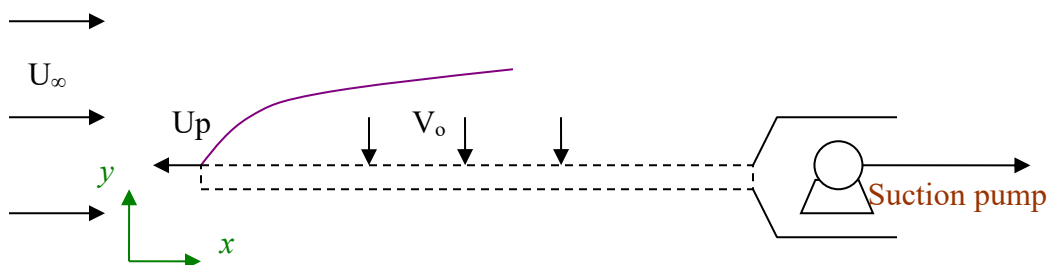
Due date: Thursday, October 19, 2023, during lecture. Submit as a hardcopy.

PROBLEM 1

Suction through Moving, Perforated Plate

Steel plates with holes are made by continually generating sheets from a die and move with a velocity U_p , as shown below. The entire set up in this case, however, lies in a bath of moving oil, which has a velocity U_∞ as shown. Further, a pump is producing suction of the fluid through the plate holes, the magnitude of the suction velocity being given by V_o . It is of importance to calculate the boundary layer at the leading edge of the plate.

- Calculate v_x within the boundary layer assuming a quadratic polynomial profile. Mention reference frame and all boundary conditions.
- Describe (do not do it) how you would obtain $\delta(x)$. You may assume $U_\infty / V_o \gg 1$ if you need it.
- From a processing standpoint, it is desirable to remove the boundary layer. Do you think it is possible to eliminate the boundary layer under any conditions (e.g. changing U_p , U_∞ , etc.)? Be explicit in your response.



Prob 2: Rework class problem on flow around sphere with details (i.e., missed steps in lecture such as getting the 4th order differential equation) [IGNORE]

Prob 3: Rework class problem on flow around cylinder with details. [IGNORE]